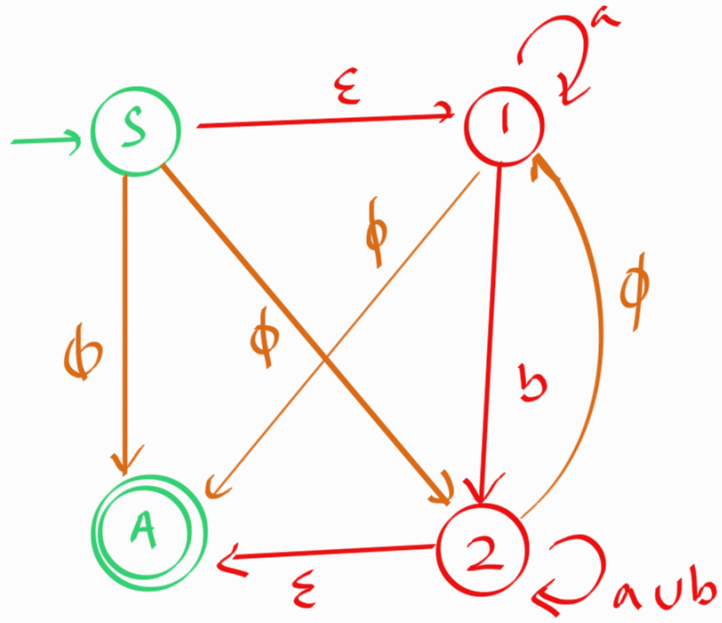
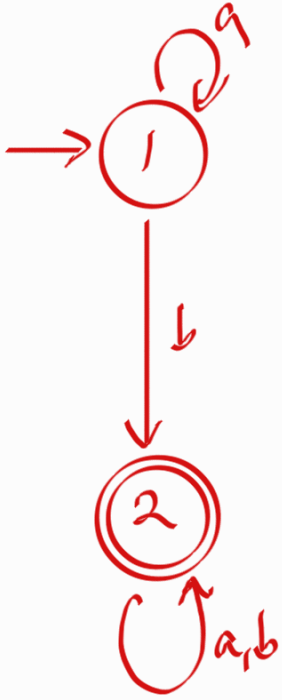


Let's look at a simple DFA
and convert it to a GNFA.



Now, we need to convert the GNFA into a regular expression.

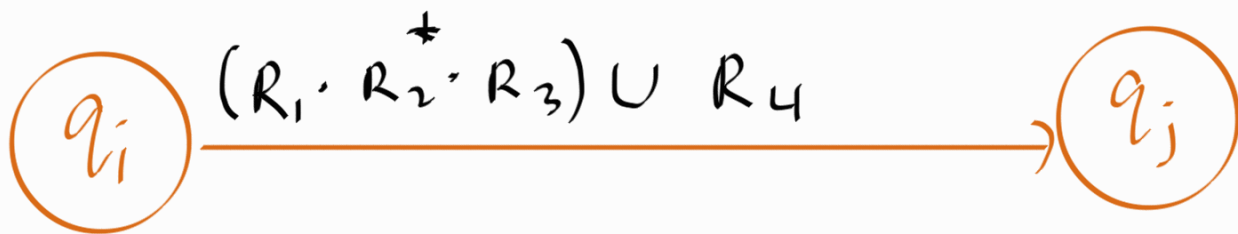
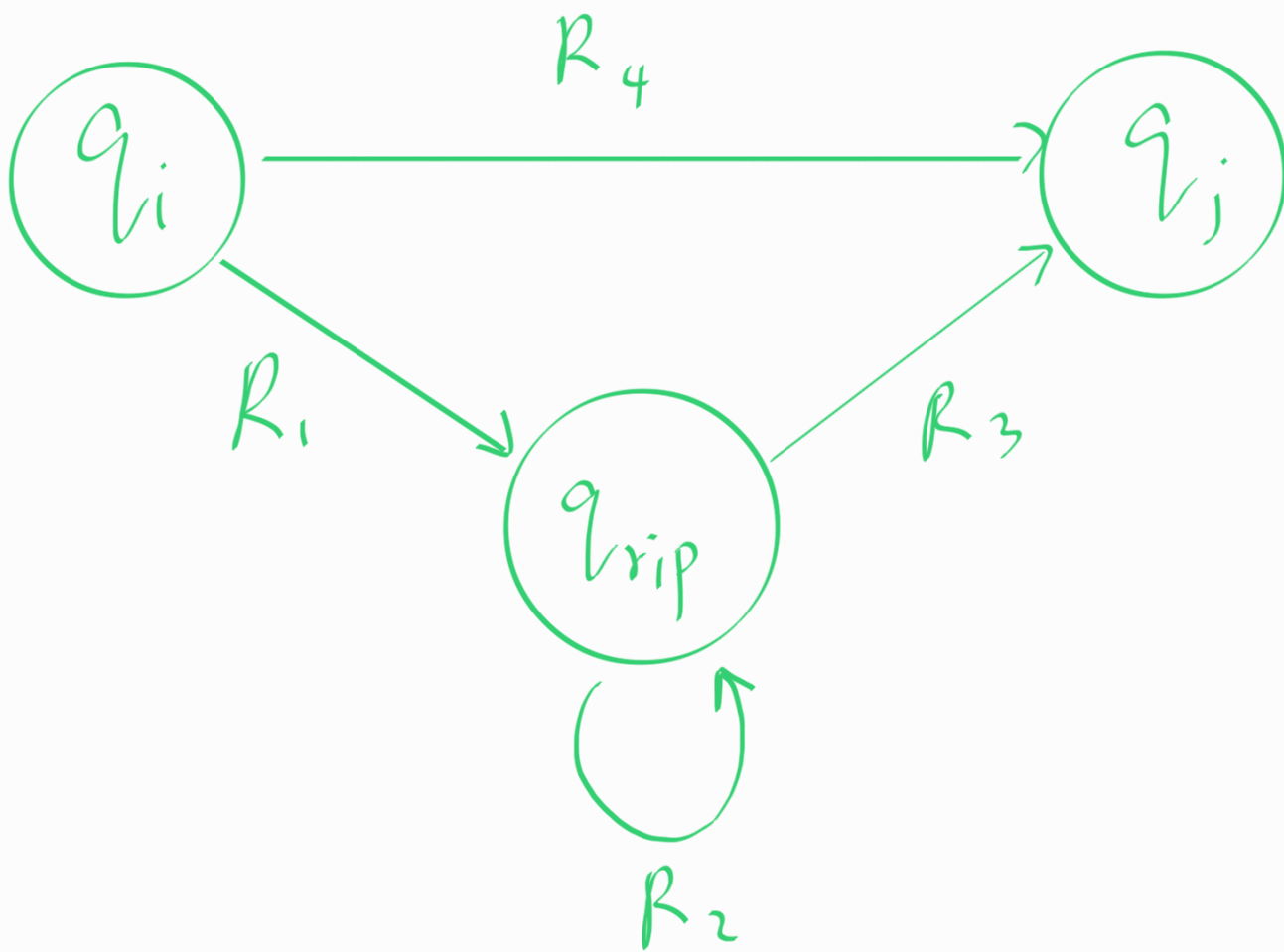
We systematically reduce the number of states of the given GNFA one by one until it has 2 states.

- Starting state
- Accept state
- Arrows between these 2 states with the final regular expression.

* The key idea about reducing the number of states is called Ripping.

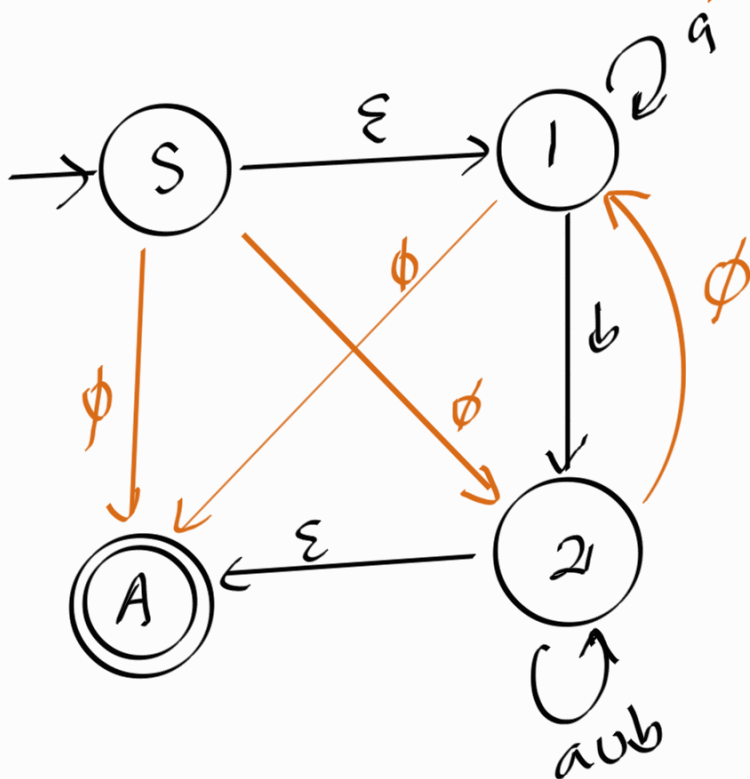
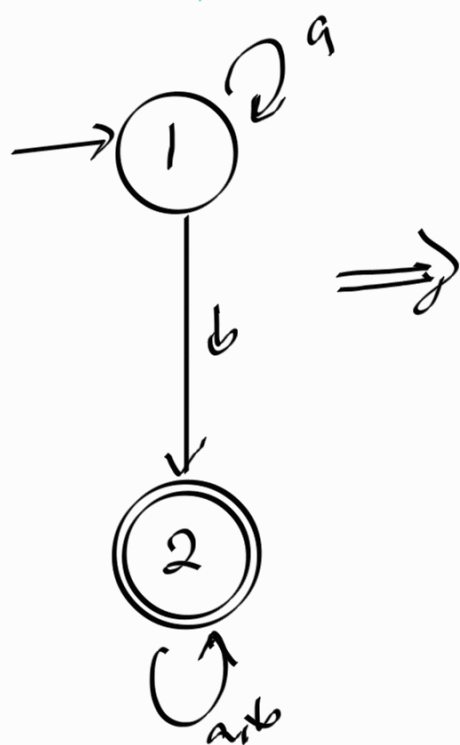
Basically, we rip a state and repair the damage, so that the same language is recognized by the GNFA.

we can rip any state except start and accept state.

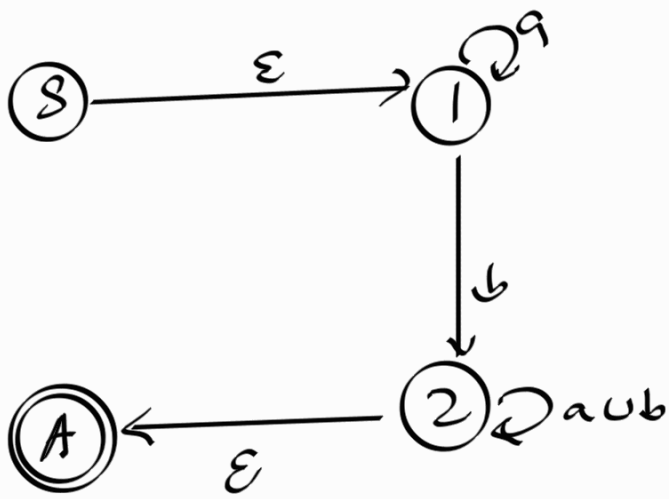


we have to consider all the paths that can go through the rip state that we consider and repair them.

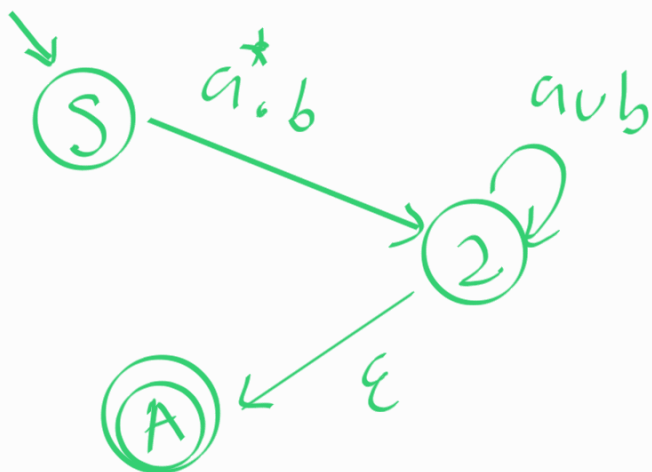
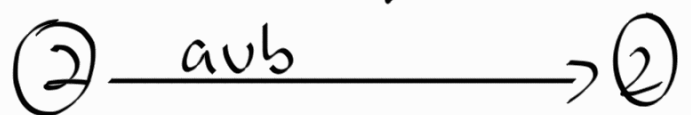
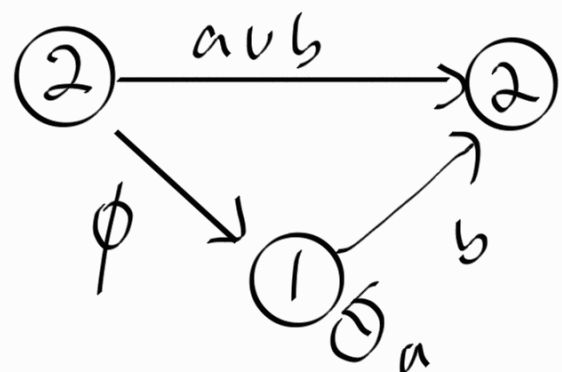
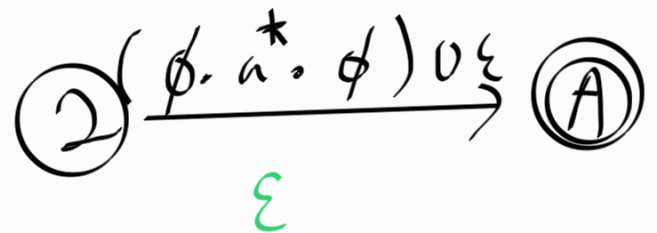
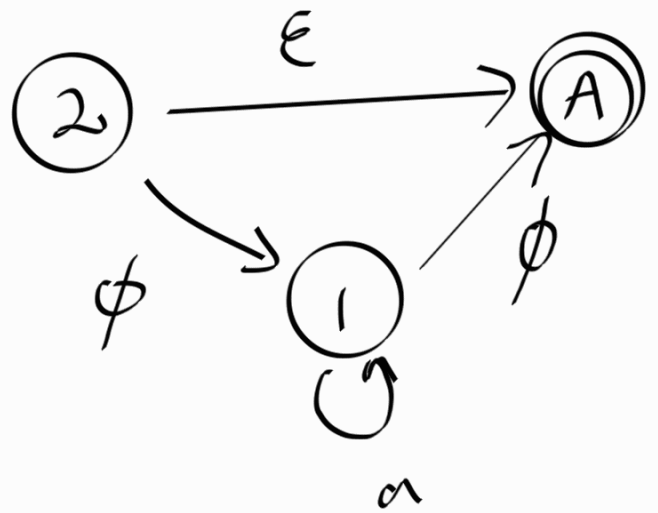
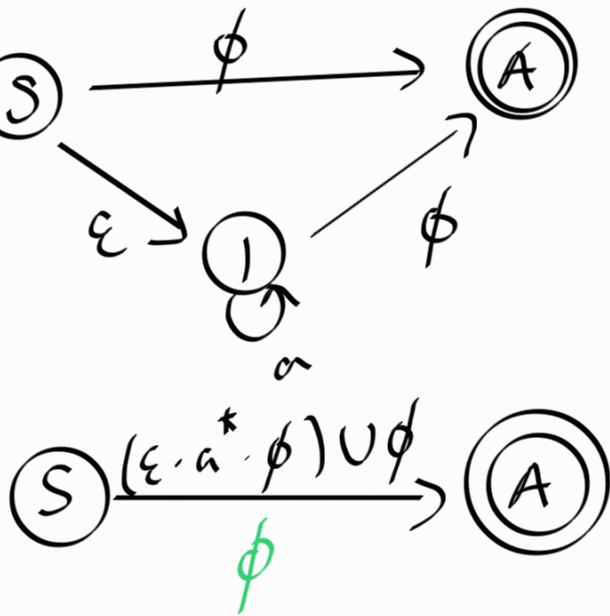
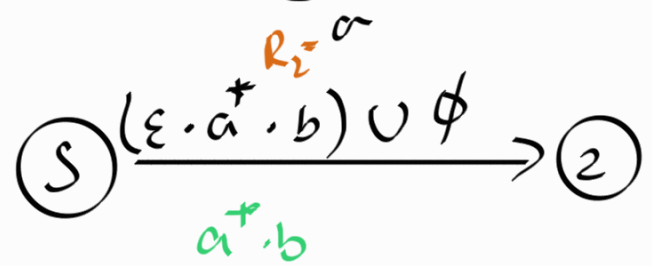
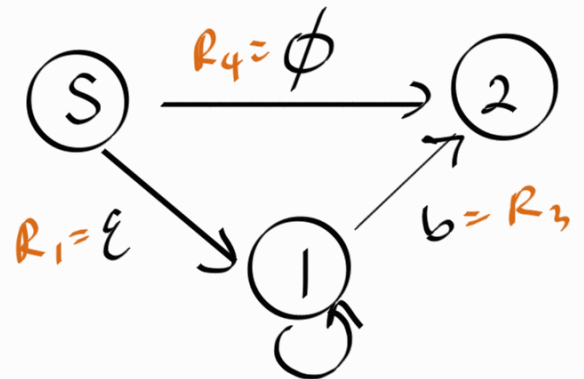
Example: DFA $\rightarrow$ GNFA $\rightarrow$ Regular Expression

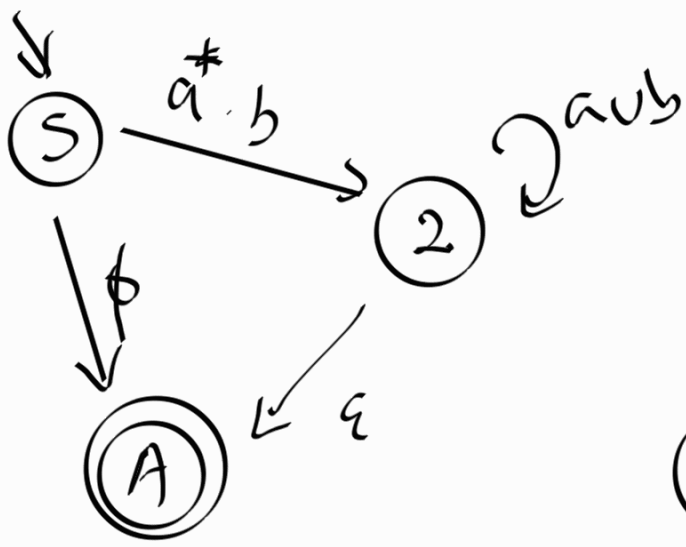


Sometimes we do not include the unnecessary ϕ transitions as it clutters the figure.



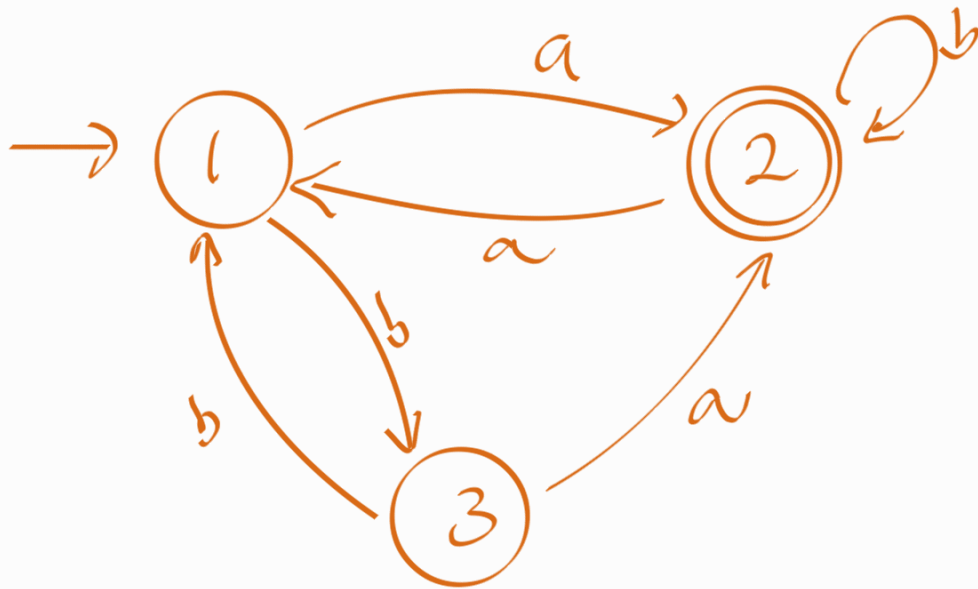
ripping ①



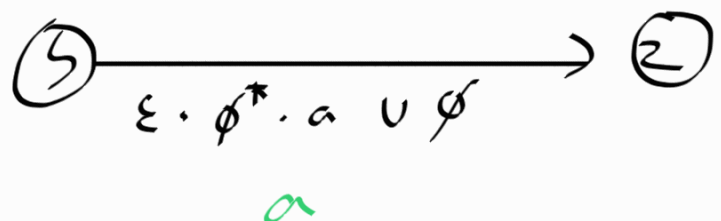
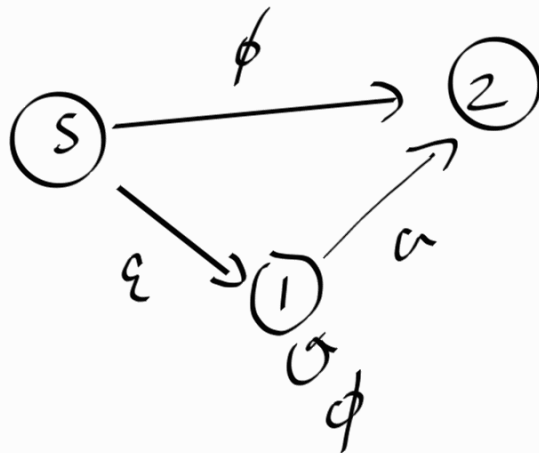
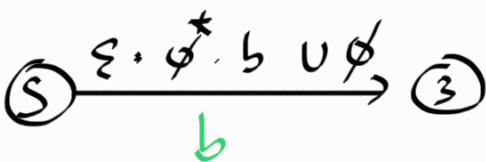
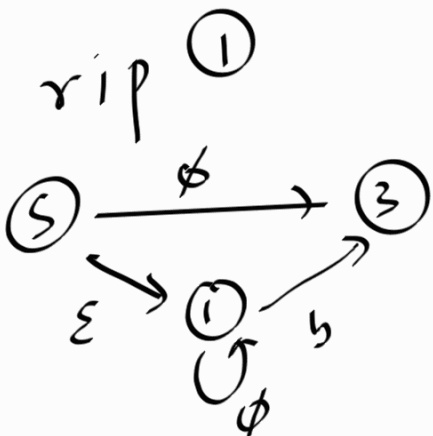
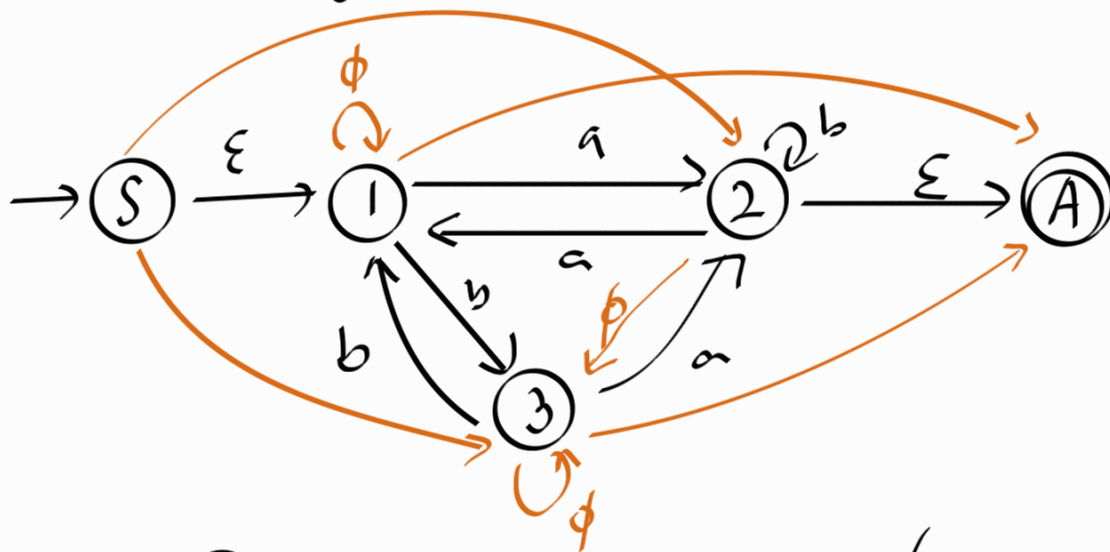


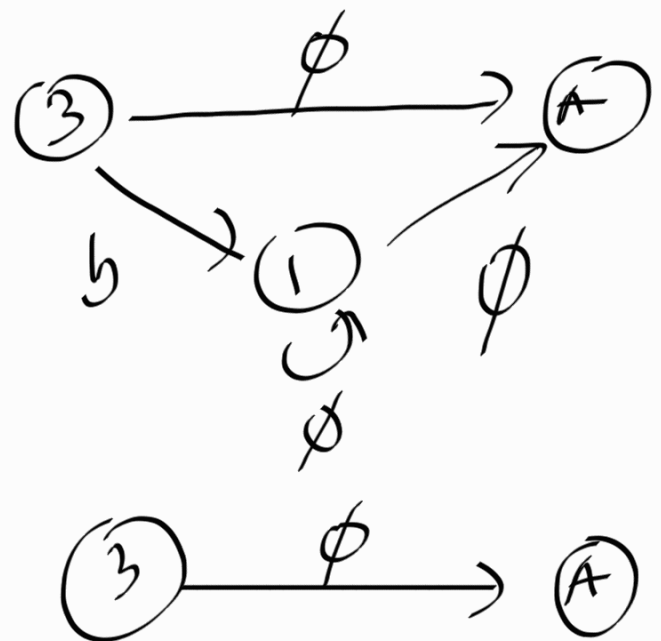
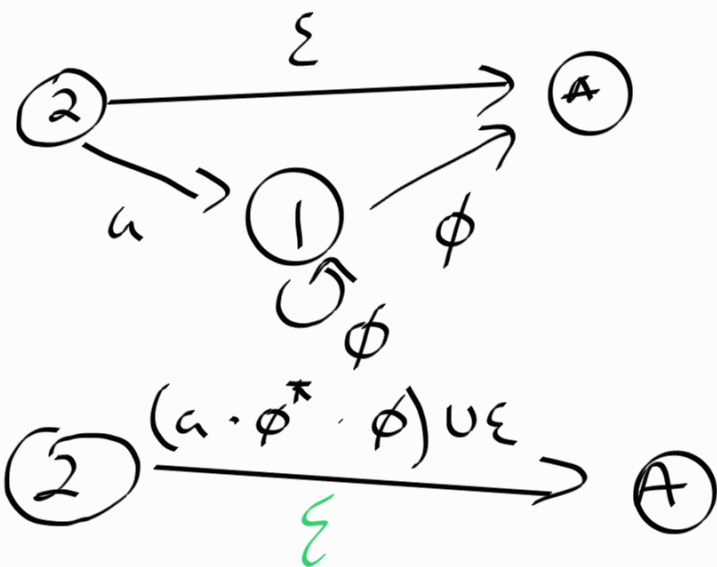
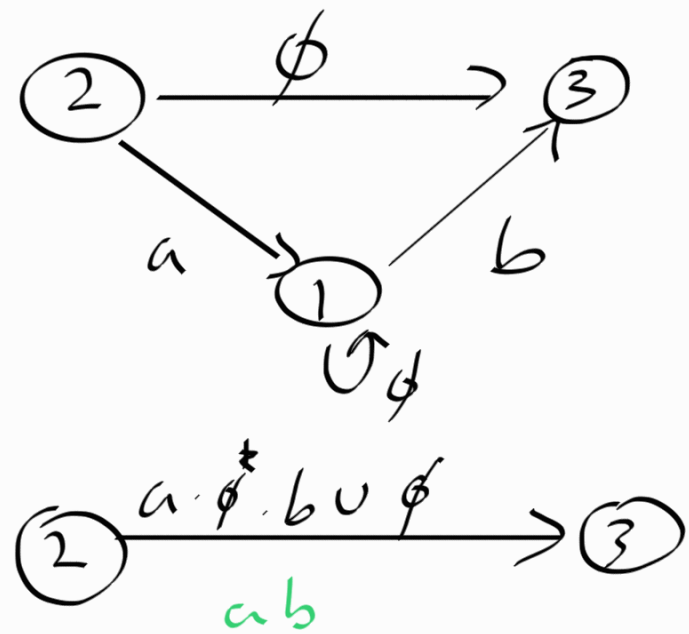
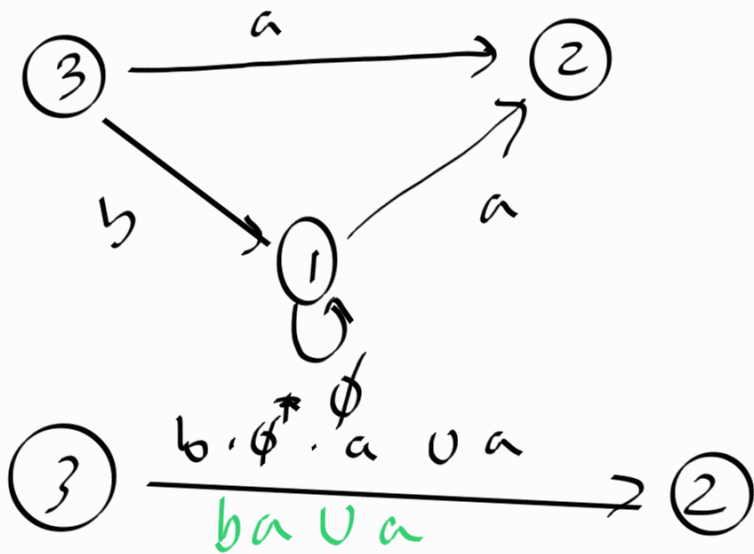
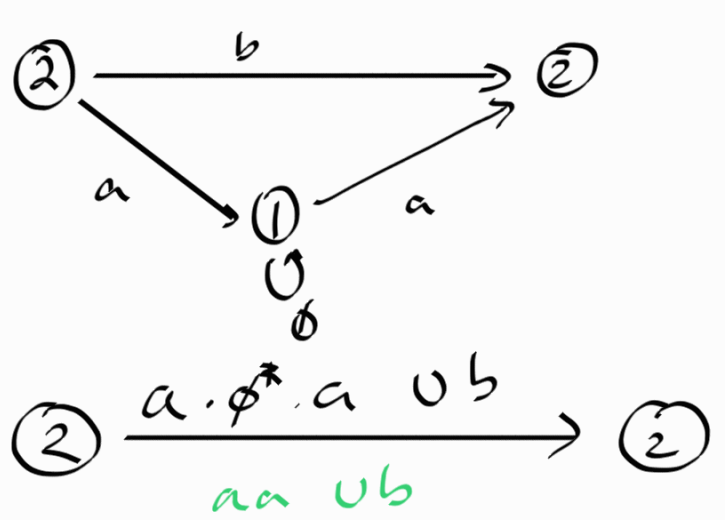
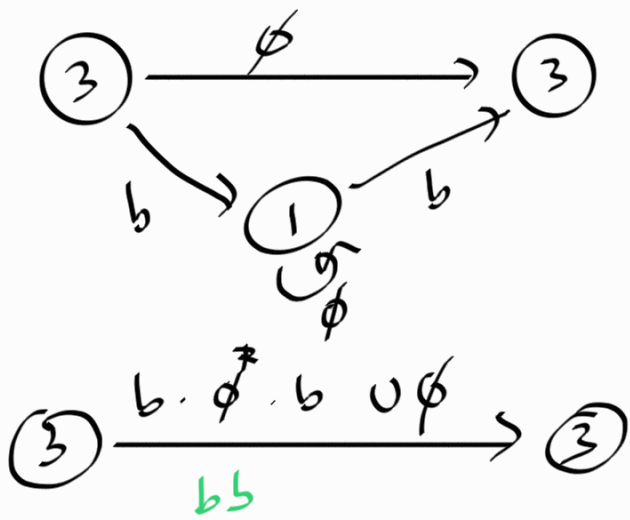
$$S \xrightarrow[(a^*.b)(a.u.b)^*]{(a^*.b).(a.u.b)^*.\epsilon) \cup \phi} A$$

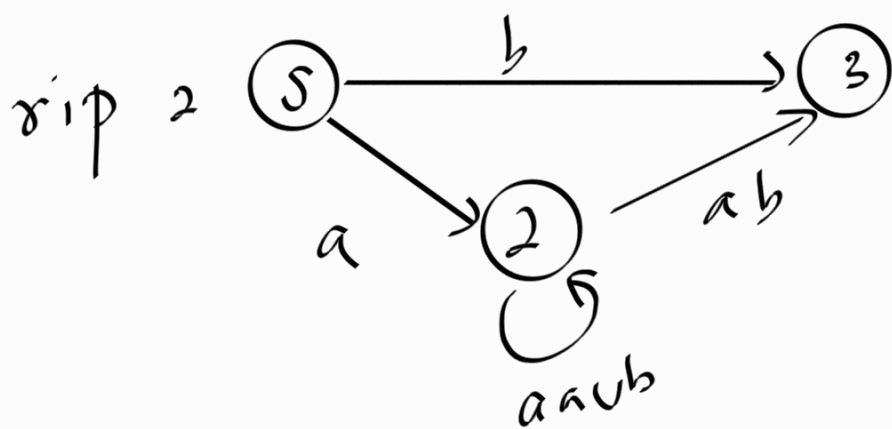
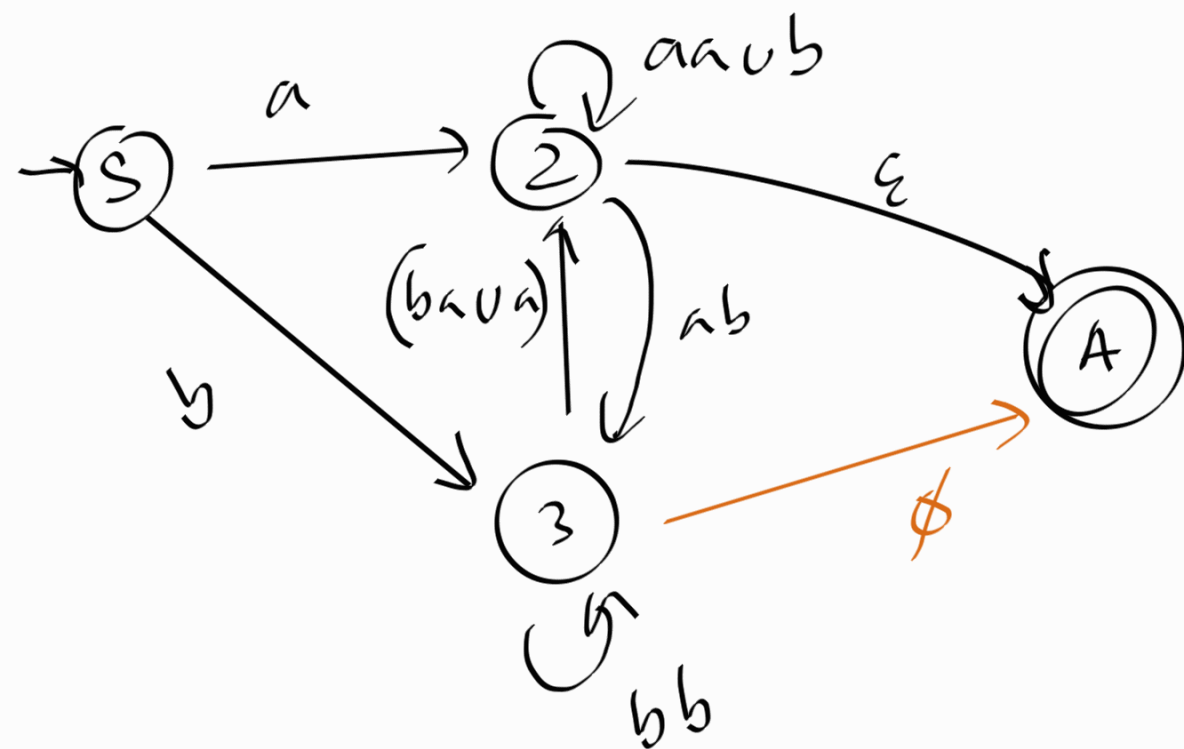
Another Example.



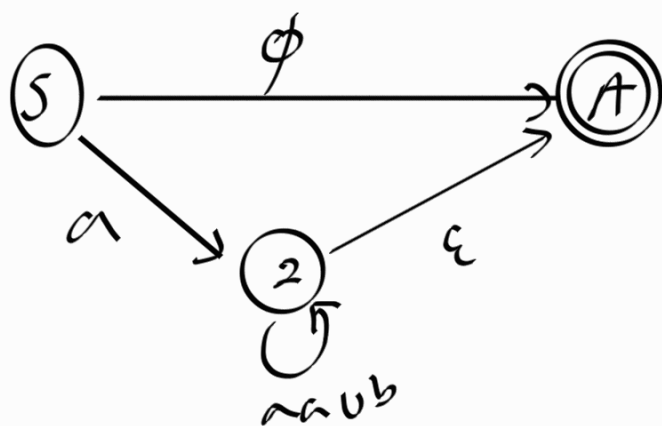
Let's try to convert to a GNFA



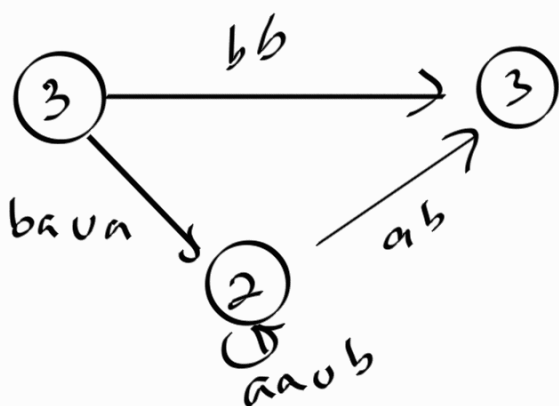




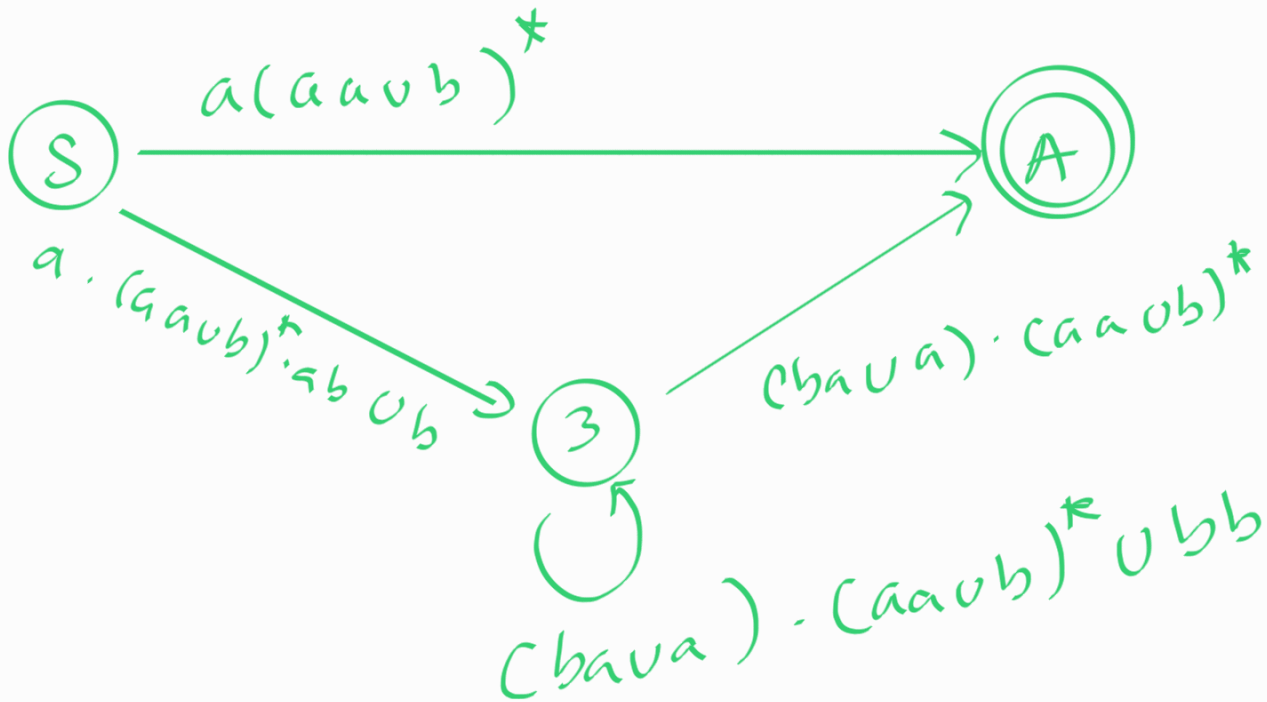
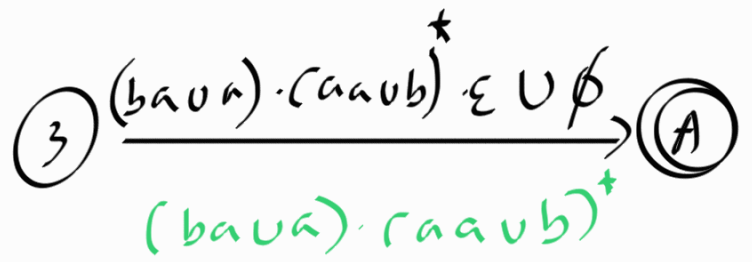
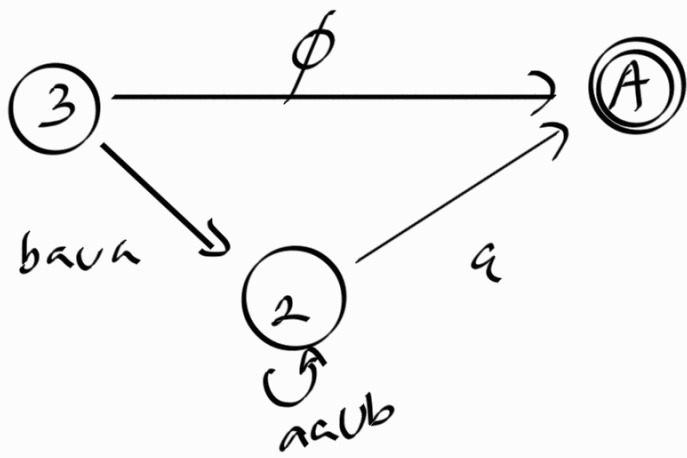
$$S \xrightarrow{a \cdot (aaub)^* \cdot ab \cup b} 3$$



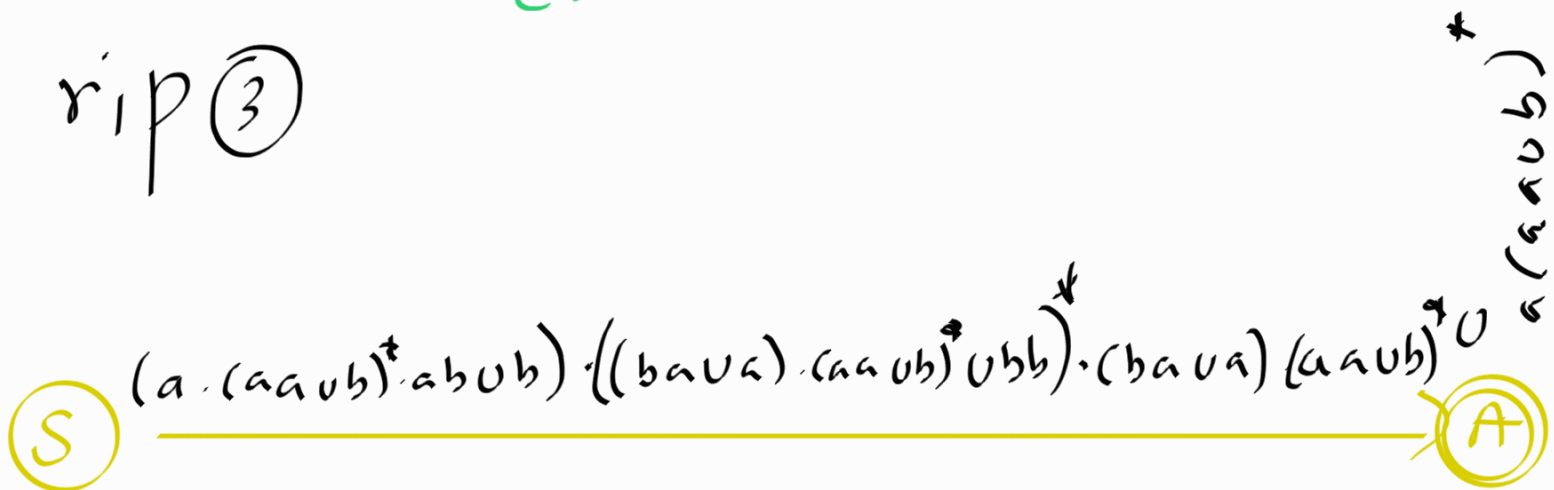
$$S \xrightarrow{\begin{matrix} a \cdot (aaub)^* \cdot \epsilon \cup \phi \\ a \cdot (aaub)^* \end{matrix}} A$$



$$3 \xrightarrow{(baub) \cdot (aaub)^* \cdot ab \cup bb} 3$$



rip ③



final regular expression.